

Telescoped Synthesis of γ -Bromo- β -Lactones from Allylic Bromides Employing Carbon Dioxide

Sreekumar Vellalath and Daniel Romo*[a]

Dedicated to Profs. Stuart L. Schreiber and K. C. Nicolaou on their receipt of the Wolf Prize

Abstract: A direct synthesis of the title compounds involving a stepwise Zn-mediated carboxylation of allylic bromides with CO₂ delivering β , γ -unsaturated carboxylic acids and a subsequent bromolactonization is reported. The described method demonstrates the use of readily prepared allylzinc bromides by the method of Knochel for fixation of CO₂ employing commodity chemicals and zinc dust. This process

was then optimized into a two-stage, telescoped process for the direct synthesis of γ -bromo- β -lactones from allyl bromides. The described strategy delivers functionalized β -lactones with dual reactivity as acylating and alkylating agents, which have utility as synthetic intermediates and are attracting growing interest as proteomic tools for activity-based protein profiling.

Keywords: allylic compounds • carbon dioxide fixation • carboxylic acids • synthetic methods • zinc

The discovery of the direct formation of allylzinc reagents from allylic bromides by Gaudemar^[1] in the early 1960s and their subsequent use in carbonyl addition reactions leading to homoallylic alcohols by Luche^[2] stand as milestones in the development of organozinc reagents for organic synthesis. These organometallic reagents offer important advantages over the corresponding lithium and magnesium counterparts, including their lower basicity, functional group tolerance, predictable reactivity, and practical preparation from zinc dust.^[3] To overcome the problematic homocoupling (Wurtz) of allyl bromides, Knochel's group introduced the use of LiCl as an additive,^[4,5] which accelerates organozinc formation. This protocol has been widely employed in carbonyl addition, acylation, and cross-coupling reactions.^[6] Although various electrophiles have been tested in combination with allylic zinc bromides generated through the Knochel protocol, to the best of our knowledge, use of the relatively unreactive electrophile CO₂ leading to β , γ -unsaturated acids has not been described.

Utilization of CO₂ as a C1 building block in organic synthesis has recently emerged as an important area of research in the context of climate change.^[7] CO₂ is a highly stable, noncombustible compound, and therefore it must typically be activated in order to convert it into synthetically useful chemicals and feedstocks.^[8] In this context, a process performed under mild reaction conditions using only atmospheric pressures of CO₂ and commercially available reagents would be highly desirable.

The carboxylation of aryl or alkyl zinc reagents catalyzed by Ni or Pd was independently reported by the Dong^[9] and Oshima^[10] groups in 2008. Subsequently, Kondo^[11] reported a transition-metal-free carboxylation

of organozinc reagents by employing a DMF/LiCl combination. Carbon–carbon bond-forming reactions employing CO₂ and nucleophilic allyl metal reagents provide a straightforward method to prepare synthetically useful β , γ -unsaturated carboxylic acids.^[12] In this regard, allyl magnesium and lithium reagents have shown lower regioselectivity and functional group tolerance,^[13] while allyl stannanes^[14] and boron reagents^[15] are useful for carboxylation with CO₂ through transition-metal catalysis. In an elegant strategy, the Martin group reported a ligand-directed regiodivergent reductive carboxylation of allyl esters with CO₂ catalyzed by Ni.^[16] Recently, Ma and Miao developed indium-mediated carboxylation of allyl halides^[17] with CO₂ under high pressure and, more recently, the same group further improved this process by employing zinc, with the best results obtained using a specific supplier of zinc powder.^[18]

Driven by our ongoing interest in β -lactone synthesis given their presence in bioactive natural products and their utility as synthetic intermediates and proteomic tools, we envisioned that a combination of allylzinc reagents and CO₂ followed by a subsequent bromolactonization would ultimately enable a telescoped three-step,

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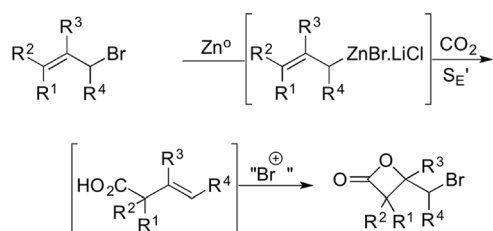


Figure 1. Direct synthesis of γ -bromo- β -lactones through a proposed telescoped carboxylation/bromolactonization process.

two-pot process for the synthesis of halogenated β -lactones directly from allyl bromides (Figure 1).

We initiated our studies with a stepwise sequence.^[19] Our initial procedure involved exposure of allylzinc bromide **2a** in THF, generated in situ from allyl bromide and activated zinc dust in the presence of LiCl according to Knochel's protocol, to a balloon (double) of CO₂ (1 atm) for 12 h. After quenching the reaction mixture with 3 N HCl, the desired, somewhat volatile acid **3a** could be isolated in 35 % yield (Table 1, entry 1). The optimization of the reaction is listed in Table 1 and included removing LiCl (Table 1, entry 2) or the substitution of LiCl with DBU (Table 1, entry 4), but this led to inferior results. Elevated temperatures (60 °C) did not improve yields (Table 1, entry 3), nor did the use of magnesium powder, which also delivered the acid but only in 22 % yield (Table 1, entry 5).

Other solvents were studied with the allylzinc bromide–LiCl complex, including DMF (Table 1, entry 6), a superior solvent for CO₂,^[20] CH₃CN (Table 1, entry 7), or toluene (Table 1, entry 8), but these gave no improvement in conversion to acid **3a**. Extended reaction times (24 h at 23 °C) with THF as solvent afforded the desired product **3a** in 52 % yield, while maintaining the CO₂

Table 1. Optimization of metal-mediated carboxylation of allyl bromide leading to 3-butenic acid (**3a**).

entry	solvent	temp. (°C)	metal (M)	additive	% yield (3)
1 ^a	THF	23	Zn ⁰	LiCl	35
2	"	"	"	-	30
3	THF	60	"	-	33
4 ^b	"	"	"	DBU	18
5	"	0 → 23	Mg ⁰	LiCl	22
6	DMF	23	Zn ⁰	"	20
7	CH ₃ CN	"	"	"	<5
8	toluene	"	"	"	12
9 ^{a,c}	THF	"	"	"	52

[a] LiCl (1.1 equiv). [b] DBU (20 mol%). [c] CO₂ in a bladder was used for 24 h.

supply at 1 atm but using a thick-walled bladder^[21] instead of a double balloon (Table 1, entry 9). We attribute the increased conversion to the ability of this bladder to slow down the exchange of CO₂ with air, which also enables extended reaction times with the same amount of CO₂.

With these conditions in hand, we briefly explored substrate scope (Table 2). The α -methyl (**1b**) and α -phenyl (**1c**) substituted allyl bromides participated in the reac-

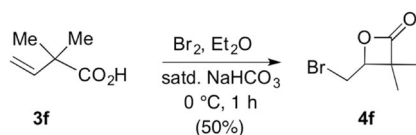
Table 2. Carboxylation of various allylic zinc bromides **2a–h** with CO₂ delivering β,γ -unsaturated carboxylic acids **3a–h**.^[a,b]

allyl bromide 1a–h	allylzinc bromide 2a–h	carboxylic acid 3a–h
3a (52% yield)	3b (56% yield)	3c (63% yield)
3e (58% yield)	3f (56% yield)	3g (46% yield)
		3h (45% yield)

[a] Reactions were performed with allylic bromides **1** (1.0 equiv, 10 mmol), zinc dust (2.0 equiv), and LiCl (1.1 equiv). [b] Yields refer to purified (flash chromatography), isolated products.

tion, leading to the corresponding β,γ -unsaturated carboxylic acids **3b** and **3c** in 56 % and 63 % yield, respectively. Interestingly, β -phenyl allyl bromide (**1d**) and β -methyl allyl bromide (**1e**) exclusively provided the S_{E'} adducts **3d** and **3e** in 37 % and 58 % yield, respectively. In the case of β -dimethyl allyl bromide (**1f**), the synthesis of acid **3f** bearing a quaternary carbon center was achieved in 56 % yield. Table 2 seems to suggest a 56 % yield for **3f**. Likewise, geranyl bromide (**1g**) participated in this reaction to provide acid **3g**, also containing a quaternary carbon, in 46 % yield. Moreover, carboxylation of a cyclic secondary allyl bromide (**1h**) afforded the cyclohexene carboxylic acid **3h** in 45 % yield. The low yields obtained upon isolation and purification of the carboxylic acids can be attributed in part to loss of the polar acid products on silica gel. However, inefficient capture of the allylzinc species by CO₂ and Wurtz coupling also led to reduced yields.

To avoid purification, we considered the possibility of a telescoped carboxylation/bromolactonization sequence. Toward this goal, we first developed conditions for bromolactonization.^[22] After a brief screening of conditions, we found that addition of molecular bromine in Et₂O at 0 °C rapidly (1 h) provided the γ -bromo- β -lactone **4f** in 50 % yield (Scheme 1).



Scheme 1. Bromo-β-lactonization of β,γ-unsaturated carboxylic acid **3f** providing bromolactone **4f**.

We then coupled the carboxylation and bromo-β-lactonization reactions with allylic bromides **1a,b, 1d–f**, and **1h** (Table 3). Some of the compound numbers in this paragraph were changed to match the information shown in

Table 3. Telescoped synthesis of bromo-β-lactones **4** using CO₂.^[a,b]

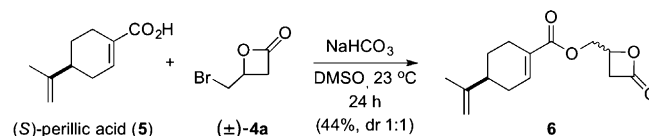
entry	allylic bromide	lactones	yield [%] (d.r.)
1			37
2			30
3		 + syn (minor)	28 (4:1)
4		 + syn (minor)	35 (4:1)
5			45
6			32

[a] Reactions were performed with 1.0 equivalents of allyl bromide **1** (10 mmol), 2.0 equivalents of zinc dust, 1.1 equivalents of LiCl, and 1.0 equivalents of Br₂. [b] Yields refer to purified, isolated β-lactones **4**.

Table 3. As before, the allylzinc reagent was generated in situ from allyl bromides **1** using zinc dust and direct carboxylation was achieved by attachment of a CO₂ bladder followed by direct quenching by addition into 3 N HCl, providing alkenoic acids **3**. Following extraction, the crude acids **3** were directly subjected to the bromination conditions described above to afford bromo-β-lactones **4**.^[22] Allyl bromide **1a** and α-methyl allyl bromide **1b** provided the corresponding β-lactones **4a** and **4b** in 37% and 30% overall yield, respectively, for the three-step process. Table 3 states yields of 37% and 30%, respectively β-phenyl (**1d**) and β-methyl (**1e**) substituted allyl bromide afforded the corresponding β-lactones **4d** and **4e** in 28% and 35% yield, respectively, as a ~4:1 mixture of diastereomers.

reomers. β-Dimethyl allyl bromide **1f** gave β-lactone **4f** containing a quaternary carbon. In a similar manner, 3-bromo-cyclohexene **1h** was converted into the corresponding bicyclic β-lactone **4h** in 32% yield.

Several β-lactone-containing natural products and derivatives are known to be covalent inhibitors of enzymes possessing catalytic nucleophilic residues (e.g., serine, threonine, or cysteine). For example, the reduced form of lipstatin, tetrahydrolipstatin (marketed as Orlistat/Alli as an anti-obesity agent), acylates an active site serine in pancreatic lipase,^[23] and the same natural product has served as a lead for novel derivatives that acylate the thioesterase domain of fatty acid synthase,^[24] and salinosporamide A potentially inhibits the 20S proteasome.^[25] β-Lactones have also been utilized more generally as proteomic tools for the covalent capture of hydrolytic enzymes.^[26] A recent example of natural product derivatization with β-lactones involves esterification of oleanic acid with a bromo-β-lactone.^[27] Additionally, the conversion of a β-hydroxy acid found in rocaglate to a β-lactone led to a bioactive natural product that covalently modified a different cellular receptor than the original natural product.^[28] We recognized the potential of utilizing the γ-bromo-β-lactones obtained in this study for natural product derivatization as a means of potentially converting non-covalent inhibitors into covalent inhibitors. As an initial demonstration, we utilized β-lactone (±)-**4a** in a base-mediated O-alkylation of perillic acid, a derivative of the essential oil perillyl alcohol (Scheme 2). Use of NaHCO₃ in DMSO with bromide (±)-**4a** led to O-alkylation of the carboxylic acid to afford the desired β-lactone ester **6** in 44% yield as an inseparable 1:1 mixture of diastereomers.



Scheme 2. Utility of γ-bromo-β-lactones for natural product derivatization leading to potential activity-based probes.

In summary, utilizing Knochel's procedure for allylic zinc formation from allylic bromides, we have developed a telescoped procedure for carboxylation with CO₂ at atmospheric pressure and ambient temperature. The described method is operationally simple and scalable, and can be carried out using commercially available zinc dust. Subsequent conversion to γ-bromo-β-lactones was possible through bromolactonization. The process involved a three-step, two-pot sequence enabling the synthesis of functionalized β-lactones directly from allylic bromides. The overall route enables incorporation of CO₂ into the synthesis of β-lactones. Furthermore, the utility of functionalized bromo-β-lactones was demonstrated through

direct arming of a simple natural product with a reactive β -lactone that could enable covalent modification of a cellular target identified through activity-based protein profiling. Conversion of the described method to a flow process and applications of this methodology to further natural product derivatizations and biological screening of derived natural product β -lactone hybrids are the subject of current investigations in our laboratory.

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References

- [1] a) M. Gaudemar, *Bull. Soc. Chim. Fr.* **1962**, 974; b) M. Bellassoued, Y. Frangin, M. Gaudemar, *Synthesis* **1977**, 205–208.
- [2] J. L. Luche, J. C. Damiano, *J. Am. Chem. Soc.* **1980**, 102, 7926–7927; b) C. Petrier, J. L. Luche, *J. Org. Chem.* **1985**, 50, 910–912; c) C. Petrier, J. Einhorn, J. L. Luche, *Tetrahedron Lett.* **1985**, 26, 1449–1452.
- [3] a) P. Knochel, H. Leuser, L.-Z. Cong, S. Perrone, F. F. Kneisel, in *Handbook of Functionalized Organometallics* (Ed.: P. Knochel), Wiley-VCH, Weinheim, **2005**, Vol. 1, Chapter 7, pp. 251–346; b) *Organozinc Reagents* (Eds.: P. Knochel, P. Jones), Oxford University Press, New York, **1999**.
- [4] A. Krasovskiy, V. Malakhov, A. Gavryushin, P. Knochel, *Angew. Chem. Int. Ed.* **2006**, 45, 6040–6044.
- [5] For a recent study on the role of LiCl additive, see: C. Feng, D. W. Cunningham, Q. T. Easter, S. A. Blum, *J. Am. Chem. Soc.* **2016**, 138, 11156–11159.
- [6] a) H. Ren, G. Dunet, P. Mayer, P. Knochel, *J. Am. Chem. Soc.* **2007**, 129, 5376–5377; b) M. D. Helm, P. Mayer, P. Knochel, *Chem. Commun.* **2008**, 1916–1917; c) C. Sämann, P. Knochel, *Synthesis* **2013**, 1870–1876; d) M. Ellwart, P. Knochel, *Angew. Chem. Int. Ed.* **2015**, 54, 10662–10665.
- [7] For recent reviews on CO₂ incorporation reactions, see: a) T. Sakakura, J.-C. Choi, H. Yasuda, *Chem. Rev.* **2007**, 107, 2365–2387; b) M. Mori, *Eur. J. Org. Chem.* **2007**, 4981–4993; c) A. Correa, R. Martin, *Angew. Chem. Int. Ed.* **2009**, 48, 6201–6204; d) S. N. Riduan, Y. Zhang, *Dalton Trans.* **2010**, 39, 3347–3357; e) R. Martin, A. W. Kleij, *ChemSusChem* **2011**, 4, 1259–1263; f) M. Peters, B. Köhler, W. Kuckshinrichs, W. Leitner, P. Markewitz, T. E. Müller, *ChemSusChem* **2011**, 4, 1216–1240; g) M. Cokoja, C. Bruckmeier, B. Rieger, W. A. Herrmann, F. E. Kühn, *Angew. Chem. Int. Ed.* **2011**, 50, 8510–8537; h) I. I. F. Boogaerts, S. P. Nolan, *Chem. Commun.* **2011**, 47, 3021–3024; i) Y. Tsuji, T. Fujihara, *Chem. Commun.* **2012**, 48, 9956–9964; j) X. Cai, B. Xie, *Synthesis* **2013**, 45, 3305–3324; k) C. Maeda, Y. Miyazaki, T. Ema, *Catal. Sci. Technol.* **2014**, 4, 1482–1497; l) B. Yu, L.-N. He, *ChemSusChem* **2015**, 8, 52–62; m) M. Aresta, A. Dibenedetto, A. Angelini, *Chem. Rev.* **2014**, 114, 1709–1742.
- [8] a) A. Decortes, A. M. Castilla, A. W. Kleij, *Angew. Chem. Int. Ed.* **2010**, 49, 9822–9837; b) M. North, R. Pasquale, C. Young, *Green Chem.* **2010**, 12, 1514–1539; c) Z.-Z. Yang, L.-N. He, J. Gao, A.-H. Liu, B. Yu, *Energy Environ. Sci.* **2012**, 5, 6602–6639; d) N. Kielland, C. J. Whiteoak, A. W. Kleij, *Adv. Synth. Catal.* **2013**, 355, 2115–2138; e) Q. Liu, L. Wu, R. Jackstell, M. Beller, *Nat. Commun.* **2015**, 6, 5933; f) P. P. Pescarmona, M. Taherimehr, *Catal. Sci. Technol.* **2012**, 2, 2169–2187; g) F. Juliá-Hernández, M. Gaydou, E. Serrano, M. van Gemmeren, R. Martin, *Top. Curr. Chem.* **2016**, 374, 45.
- [9] C. S. Yeung, V. M. Dong, *J. Am. Chem. Soc.* **2008**, 130, 7826–7827.
- [10] H. Ochiai, M. Jang, K. Hirano, H. Yorimitsu, K. Oshima, *Org. Lett.* **2008**, 10, 2681–2683.
- [11] K. Kobayashi, Y. Kondo, *Org. Lett.* **2009**, 11, 2035–2037.
- [12] For some applications of β,γ -unsaturated carboxylic acids in synthesis, see: a) A. G. Myers, D. R. Siegel, D. J. Buzard, M. G. Charest, *Org. Lett.* **2001**, 3, 2923–2926; b) P. R. Andreada, J. S. McLellan, Y. Chen, P. G. Wang, *Org. Lett.* **2002**, 4, 3875–3878; c) T. Fekner, H. Muller-Bunz, P. J. Guiry, *Org. Lett.* **2006**, 8, 5109–5112; d) P. D. O'Connor, U. B. Kim, M. A. Brimble, *Eur. J. Org. Chem.* **2009**, 4405–4411; e) T. den Hartog, B. Macia, A. J. Minnaard, B. L. Feringa, *Adv. Synth. Catal.* **2010**, 352, 999–1013; f) Z. Gu, A. Zakarian, *Org. Lett.* **2011**, 13, 1080–1082.
- [13] a) R. T. Sanderson, *J. Am. Chem. Soc.* **1955**, 77, 4531–4532; b) L. E. Friedrich, R. Cormier, *J. Org. Chem.* **1971**, 36, 3011–3015; c) G. Courtois, L. Miginiac, *J. Organomet. Chem.* **1974**, 69, 1–44; d) W. Oppolzer, E. P. Kundig, P. M. Bishop, C. Perret, *Tetrahedron Lett.* **1982**, 23, 3901–3904; e) G. Zadel, E. Breitmaier, *Angew. Chem. Int. Ed.* **1992**, 31, 1035–1036.
- [14] M. Shi, K. M. Nicholas, *J. Am. Chem. Soc.* **1997**, 119, 5057–5058.
- [15] H. A. Duong, P. B. Huleatt, Q.-W. Tan, E. L. Shuying, *Org. Lett.* **2013**, 15, 4034–4037.
- [16] T. Moragas, J. Cornella, R. Martin, *J. Am. Chem. Soc.* **2014**, 136, 17702–17705.
- [17] B. Miao, S. Ma, *Chem. Commun.* **2014**, 50, 3285–3287.
- [18] The procedure by Ma utilizes zinc dust from Sino Pharm, which is not available in the United States: B. Miao, S. Ma, *Org. Chem. Front.* **2015**, 2, 65–68.
- [19] For a review on bromolactonization, see: S. Robin, G. Rousseau, *Eur. J. Org. Chem.* **2002**, 3099–3114.
- [20] For the solubility of CO₂ in various solvents, see: G. R. Pazuki, H. Pahlavanzadeh, *Theor. Found. Chem. Eng.* **2005**, 39, 240–245.
- [21] The heavy-duty rubber balloons (punching bag bladders) were purchased from Amazon (see Supporting Information for details).
- [22] a) W. E. Barnett, J. C. McKenna, *Tetrahedron Lett.* **1971**, 23, 2595–2598; b) I. Shibata, M. Toyota, A. Baba, H. Matsuda, *J. Org. Chem.* **1990**, 55, 2487–2491.
- [23] C. Lowe, J. C. Vederas, *Org. Prep. Proced. Int.* **1995**, 27, 305–346.
- [24] a) S. J. Kridel, F. Axelrod, N. Rozenkrantz, J. W. Smith, *Cancer Res.* **2004**, 64, 2070–2075; b) L. M. Knowles, F. Axelrod, C. D. Browne, J. W. Smith, *J. Biol. Chem.* **2004**, 279, 30540–30545; c) W. Zhang, R. D. Richardson, S. Chamni, J. W. Smith, D. Romo, *Bioorg. Med. Chem. Lett.* **2008**, 18, 2491–2494.
- [25] R. H. Feling, G. O. Buchanan, T. J. Mincer, C. A. Kauffman, P. R. Jensen, W. Fenical, *Angew. Chem. Int. Ed.* **2003**, 42, 355–357.
- [26] a) T. Bottcher, S. A. Sieber, *Angew. Chem. Int. Ed.* **2008**, 47, 4600–4603; b) M. Groll, R. Huber, B. C. M. Potts, *J. Am. Chem. Soc.* **2006**, 128, 5136–5141; c) E. Zeiler, A. List, F.

- Alte, M. Gersch, R. Wachtel, M. Poreba, M. Drag, M. Groll, S. A. Sieber, *Proc. Natl. Acad. Sci.* **2013**, *110*, 11302–11307; d) T. Böttcher, S. A. Sieber, *Angew. Chem. Int. Ed.* **2007**, *47*, 4600–4603.
- [27] a) T. Chen, M. Tong, K. Ma, D. Han, L. Zhang, Patent CN 105017374A, Nov 04, **2015**; b) T. Chen, M. Tong, K. Ma, L. Zhang, X. Xi, *Nat. Prod. Res. Dev.* **2016**, *28*, 11–16.
- [28] N. J. Lajkiewicz, A. B. Coggins, III, M. J. Niphakis, B. F. Cravatt, J. A. Porco, Jr., *J. Am. Chem. Soc.* **2014**, *136*, 2659–2664.

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